

Course Project: Item-Level Price Co-Movement Network using CPI Data

Discrete Structures

1. Project Objective

The objective of this project is to model and analyze the similarity of consumer items based on their price movement patterns across multiple cities. Instead of representing cities as graph nodes, this project represents **items (products) as nodes**. Two items are considered adjacent if a large number of cities exhibit similar price increase/decrease patterns for the two items.

Students will apply concepts of relations, graphs, similarity measures, and centrality metrics to construct and analyze an item-item network. The project also includes a temporal component using three years of monthly CPI data.

2. Data Description

The CPI data is provided on a monthly basis for multiple cities and multiple consumer items.

The dataset spans **three consecutive years**. Each record contains the price of an item in a specific city and month.

Students must use the following dataset:

<https://www.pbs.gov.pk/price/>

3. Node and Feature Representation

Let:

- I be the set of items,
- C be the set of cities,
- T_y be the set of months of year y .

Each item is treated as a node in the network.

For a fixed city c and item i , students must compute a **price-change pattern vector** for each year y :

$$\mathbf{v}_{i,c}^{(y)} = (\Delta p_{i,c}^{(y,2)}, \Delta p_{i,c}^{(y,3)}, \dots, \Delta p_{i,c}^{(y,12)}),$$

where $\Delta p_{i,c}^{(y,m)}$ denotes the price change from month $m - 1$ to month m .

4. Item Similarity within a City

For two items i and j in a city c and year y , students must compute the cosine similarity:

$$\text{sim}_c^{(y)}(i, j) = \frac{\mathbf{v}_{i,c}^{(y)} \cdot \mathbf{v}_{j,c}^{(y)}}{\|\mathbf{v}_{i,c}^{(y)}\| \|\mathbf{v}_{j,c}^{(y)}\|}.$$

This measures how similar the monthly price-change patterns of the two items are within that city for that year.

5. Aggregation across Cities

For a fixed year y , define

$$N_y(i, j) = \left| \left\{ c \in C \mid \text{sim}_c^{(y)}(i, j) \geq \tau \right\} \right|,$$

where τ is a similarity threshold.

6. Graph Construction for a Single Year

For each year y , construct an undirected graph

$$G_y = (I, E_y),$$

where

$$(i, j) \in E_y \iff N_y(i, j) \geq K,$$

and K is a user-defined city-count threshold.

Hence, two items are connected if a sufficiently large number of cities show similar price movement patterns for those items in that year.

7. Weighted Graph Variant

Students must also construct a weighted version of the network where

$$w_y(i, j) = N_y(i, j),$$

or alternatively

$$w_y(i, j) = \frac{1}{|C|} \sum_{c \in C} \text{sim}_c^{(y)}(i, j).$$

8. Year-wise Pattern Analysis

Students must construct and analyze a separate graph for each of the three years:

$$G_{y_1}, G_{y_2}, G_{y_3}.$$

For each yearly graph, students must compute and interpret:

- degree centrality,
- closeness centrality,
- betweenness centrality.

Students must identify highly central items and interpret their role in terms of price co-movement.

9. Temporal Network Analysis across Three Years

Let the three yearly graphs be ordered in time.

Students must study:

- appearance and disappearance of edges across years,
- stability of highly central items across years,
- changes in connected components or clusters of items.

A temporal comparison must be performed to identify persistent and transient item relationships.

10. Category-based Extension

Each item belongs to one of the predefined seven categories. Students must analyze:

- connectivity within categories,
- connectivity between categories,
- whether clusters of items align with category boundaries.

11. Threshold and Weight Sensitivity Analysis

Students must analyze the effect of:

- varying the similarity threshold τ ,
- varying the city-count threshold K ,
- varying the weighting schemes for edges.

The resulting changes in graph structure and centrality rankings must be clearly reported.

12. Required Implementation

Students must implement the complete pipeline:

1. data preprocessing,
2. construction of yearly item similarity matrices,
3. construction of yearly graphs,
4. computation of centrality measures,
5. temporal comparison across three years.

The implementation must be reproducible.

13. Demo Video Requirement

Students must produce a demonstration video of **maximum duration 8 minutes**. The video should be carefully structured to reflect both the technical depth and the analytical outcomes of the project.

- **Methodology and Implementation (at least 5 minutes):**
 - explanation of data preprocessing and feature construction,
 - description of similarity computation and thresholding,
 - details of graph construction and weighting schemes,
 - implementation choices, tools, and libraries used.
- **Results and Analysis (up to 3 minutes):**
 - visualization of at least two different weighting schemes,
 - comparison of graphs for different thresholds,
 - discussion of the effect of cosine similarity thresholds,
 - changes in centrality rankings under different weights,
 - key temporal insights across the three yearly networks.

The video must be uploaded to YouTube and the link must be provided in the report.

Important Note: Each group will also deliver a **formal in-class presentation**. The demo video is an **additional requirement** and does not replace the presentation. The presentation will be used for interactive evaluation, while the video will serve as a concise and well-prepared technical summary of the project.

14. Deliverables

Each group must submit:

- a technical report describing methodology and results,
- source code used for the analysis,
- a YouTube link to the demonstration video.

15. Evaluation Rubric

The course project will be evaluated out of **100 marks** based on the following components:

Criterion	Marks	Description
Problem Understanding and Formulation	10	Clarity in defining the problem, objectives, and scope. Proper interpretation of requirements and assumptions.
Methodology and Design	15	Soundness of approach, justification of techniques (e.g., similarity measures, graph modeling), and logical organization of steps.
Implementation and Technical Understanding	40	Correct and efficient implementation, appropriate use of tools/libraries, and adherence to good coding practices.
Visualization of Results and Analysis	20	Quality and clarity of graphs and plot, Depth of analysis, comparison across different configurations (e.g., thresholds, weights), and ability to draw meaningful insights.
Report Quality	10	Organization, clarity of writing, completeness, and proper formatting. Logical flow and coherence of arguments.
Demo Video	5	Adherence to time constraints, clarity in explaining methodology and results, and overall quality of presentation.
Total	100	

Note:

- Greater weight is assigned to **methodology and implementation**, reflecting the importance of technical depth.
- Superficial work with strong presentation will not receive high marks without sound analysis and correctness.
- Originality, proper attribution, and adherence to academic integrity policies are mandatory.